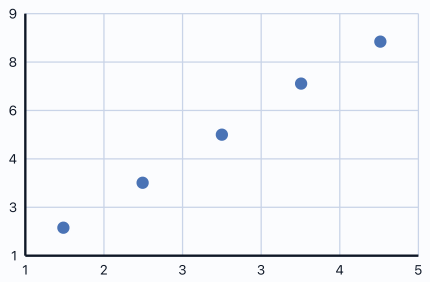
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Combinatorics, Binomial Probability & Regression Practice Exercises

1 MCQ

The graph of the function $f(x) = (x-1)^2 - 4$ is shown in the standard (x, y) coordinate plane below. What is the range of f?



A $[-4, \infty)$
B $[1, \infty)$
C $(-\infty, -4]$
D $(-\infty, \infty)$

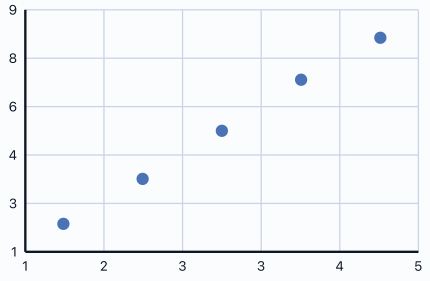
2 MCQ

For the complex number $i = \sqrt{-1}$, which of the following is equivalent to the expression $(5 - 3i) - (2 - 7i)$?

A $3 + 4i$
B $3 - 10i$
C $7 + 4i$
D $3 - 4i$

3 MCQ

What is the value of $\lim_{x \to 3} \frac{x^2 - 2x - 3}{x - 3}$?



A 4
B 0
C 6
D The limit does not exist

4 MCQ

Given the vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$, what is the vector $2\mathbf{u} + \mathbf{v}$?

A $\langle 4, -3 \rangle$

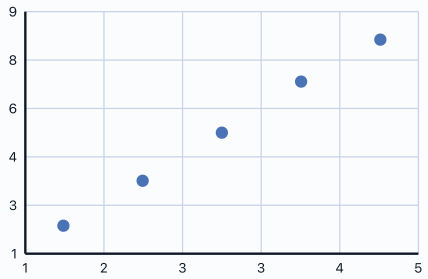
B $\langle 1, 1 \rangle$

C $\langle 4, 1 \rangle$

D $\langle 8, -13 \rangle$

5 MCQ

The first four terms of an arithmetic sequence are 5, 12, 19, and 26. What is the 20th term of this sequence?



A 131

B 138

C 140

D 145

6 MCQ

A point has polar coordinates $(r, \theta) = (6, \frac{2\pi}{3})$. What are the Cartesian coordinates (x, y) of this point?

A $(-3, 3\sqrt{3})$

B $(-3\sqrt{3}, 3)$

C $(3, -3\sqrt{3})$

D $(3\sqrt{3}, -3)$

7 MCQ

The polynomial function $P(x) = 2x^3 + ax^2 - 5x + b$ has $(x - 1)$ as a factor. When $P(x)$ is divided by $(x + 2)$, the remainder is -18. What is the remainder when $P(x)$ is divided by $(x - 3)$?

A 2

B 8

C 18

D -30

8 MCQ

For what real value of k is the complex expression

$$\frac{k+8i}{3-i}+\frac{2-ki}{3+i}$$

a real number?

A 11

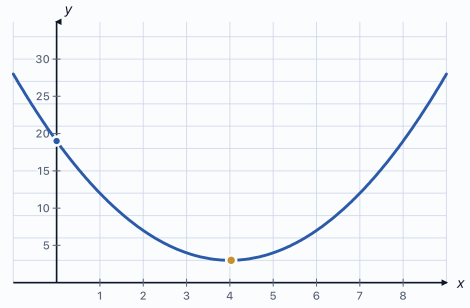
B 1

C 6

D -11

9 MCQ

In Conics, 3D Geometry & Probability (problem set #925), consider the function $f_{925}(x)=x^2-8x+19$ illustrated on the coordinate axes. Express $f_{925}(x)$ in the completed-square form $(x-h)^2+k$ and evaluate $k+f_{925}(6)$.



A 8

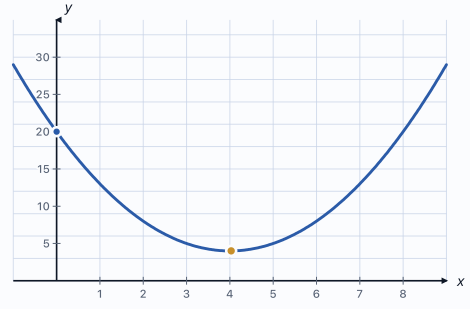
B 14

C 7

D 10

10 MCQ

Considering Conics, 3D Geometry & Probability (problem set #926), consider the function $f_{926}(x)=x^2-8x+20$ illustrated on the coordinate axes. Express $f_{926}(x)$ in the completed-square form $(x-h)^2+k$ and evaluate $k+f_{926}(7)$.



A 21

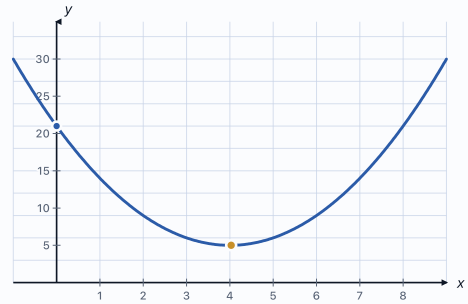
B 13

C 17

D 14

11 MCQ

Examining Conics, 3D Geometry & Probability (problem set #927), consider the function $f_{927}(x) = x^2 - 8x + 21$ illustrated on the coordinate axes. Express $f_{927}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{927}(8)$.



A 30

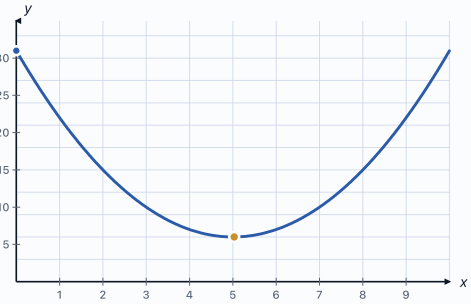
B 21

C 26

D 22

12 MCQ

For the configuration in Conics, 3D Geometry & Probability (problem set #928), consider the function $f_{928}(x) = x^2 - 10x + 31$ illustrated on the coordinate axes. Express $f_{928}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{928}(10)$.



A 42

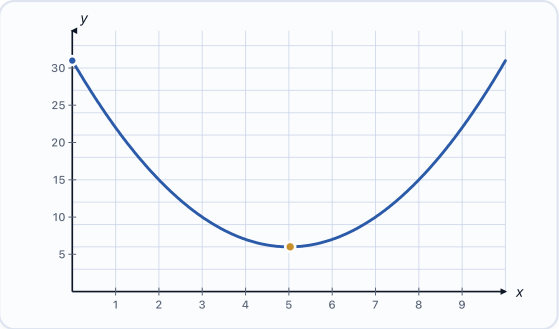
B 37

C 31

D 32

13 MCQ

Analyzing Conics, 3D Geometry & Probability (problem set #929), consider the function $f_{929}(x) = x^2 - 10x + 32$ illustrated on the coordinate axes. Express $f_{929}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{929}(11)$.



A 50

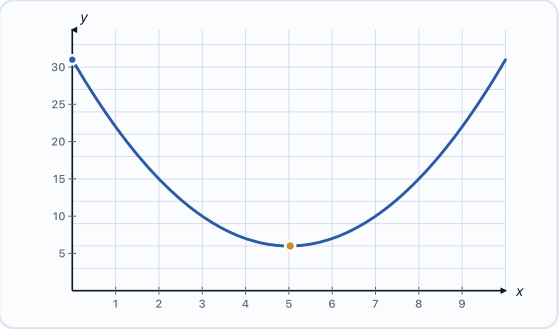
B 43

C 44

D 55

14 MCQ

Within Conics, 3D Geometry & Probability (problem set #930), consider the function $f_{930}(x) = x^2 - 10x + 33$ illustrated on the coordinate axes. Express $f_{930}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{930}(12)$.



A 70

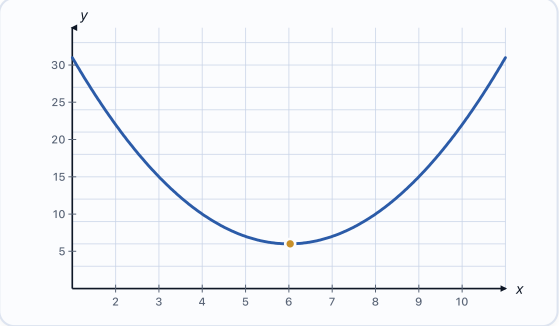
B 57

C 58

D 65

15 MCQ

Considering Conics, 3D Geometry & Probability (problem set #931), consider the function $f_{931}(x) = x^2 - 12x + 45$ illustrated on the coordinate axes. Express $f_{931}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{931}(14)$.



A 88

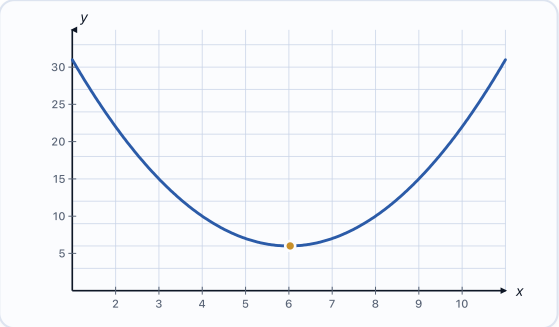
B 82

C 73

D 74

16 MCQ

Examining Conics, 3D Geometry & Probability (problem set #932), consider the function $f_{932}(x) = x^2 - 12x + 46$ illustrated on the coordinate axes. Express $f_{932}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{932}(15)$.



A 107

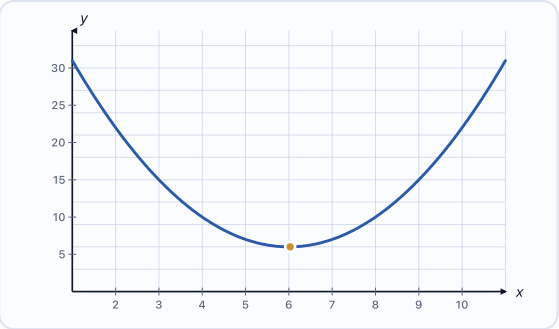
B 91

C 101

D 92

17 MCQ

For the configuration in Conics, 3D Geometry & Probability (problem set #933), consider the function $f_{933}(x) = x^2 - 12x + 47$ illustrated on the coordinate axes. Express $f_{933}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{933}(16)$.



A 122

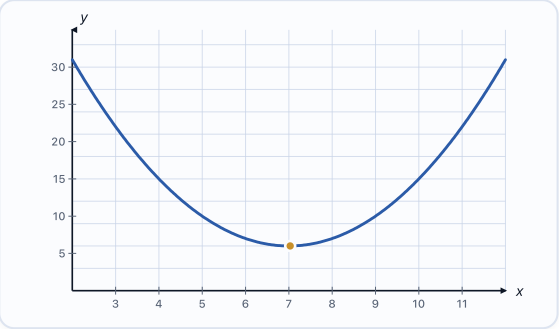
B 128

C 111

D 112

18 MCQ

Analyzing Conics, 3D Geometry & Probability (problem set #934), consider the function $f_{934}(x) = x^2 - 14x + 61$ illustrated on the coordinate axes. Express $f_{934}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{934}(9)$.



A 28

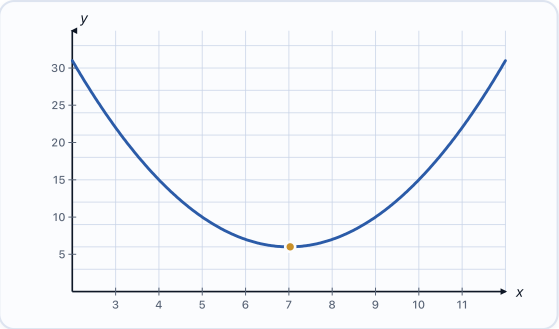
B 35

C 16

D 26

19 MCQ

Within Conics, 3D Geometry & Probability (problem set #935), consider the function $f_{935}(x) = x^2 - 14x + 62$ illustrated on the coordinate axes. Express $f_{935}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{935}(10)$.



A 42

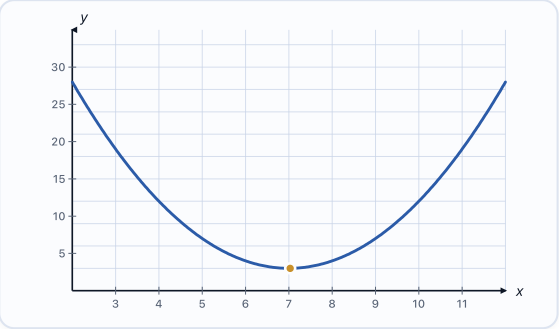
B 22

C 32

D 35

20 MCQ

Considering Conics, 3D Geometry & Probability (problem set #936), consider the function $f_{936}(x) = x^2 - 14x + 52$ illustrated on the coordinate axes. Express $f_{936}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{936}(11)$.



A 29

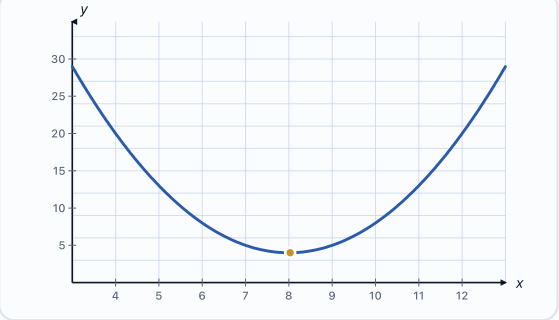
B 19

C 18

D 22

21 MCQ

Examining Conics, 3D Geometry & Probability (problem set #937), consider the function $f_{937}(x) = x^2 - 16x + 68$ illustrated on the coordinate axes. Express $f_{937}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{937}(13)$.



A 41

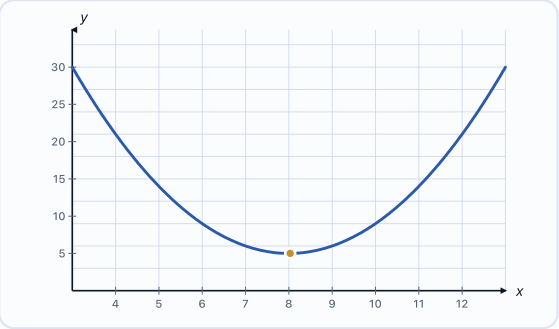
B 29

C 28

D 33

22 MCQ

For the configuration in Conics, 3D Geometry & Probability (problem set #938), consider the function $f_{938}(x) = x^2 - 16x + 69$ illustrated on the coordinate axes. Express $f_{938}(x)$ in the completed-square form $(x - h)^2 + k$ and evaluate $k + f_{938}(14)$.



A 40

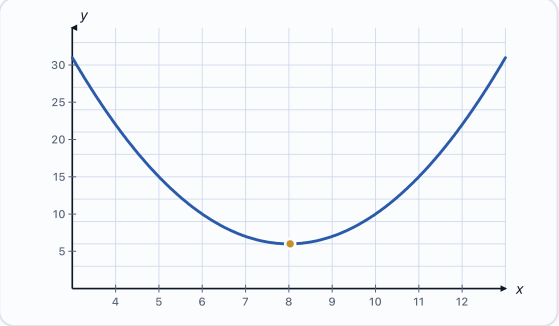
B 46

C 54

D 41

23 MCQ

Analyzing Conics, 3D Geometry & Probability (problem set #939), consider the function $f_{939}(x) = x^2 - 16x + 70$ illustrated on the coordinate axes. Express $f_{939}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{939}(15)$.



A 61

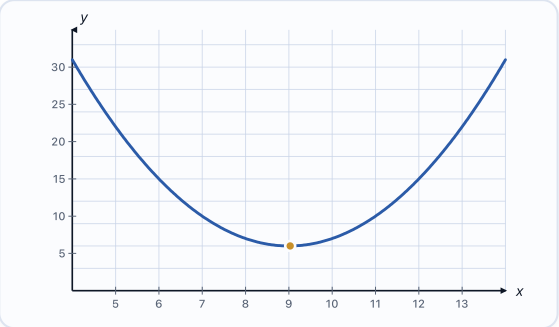
B 54

C 69

D 55

24 MCQ

Within Conics, 3D Geometry & Probability (problem set #940), consider the function $f_{940}(x) = x^2 - 18x + 88$ illustrated on the coordinate axes. Express $f_{940}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{940}(17)$.



A 71

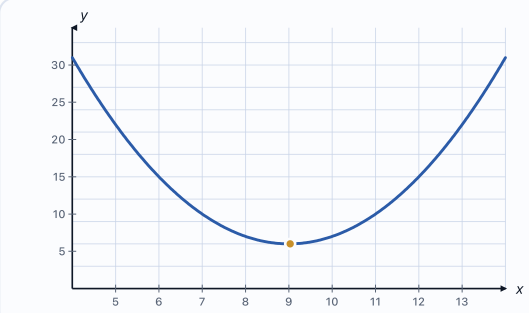
B 70

C 87

D 78

25 MCQ

Considering Conics, 3D Geometry & Probability (problem set #941), consider the function $f_{941}(x) = x^2 - 18x + 89$ illustrated on the coordinate axes. Express $f_{941}(x)$ in the completed-square form $(x-h)^2 + k$ and evaluate $k + f_{941}(18)$.



A 106

B 89

C 88

D 97